

UltraDefrag: A Choice Defragmenter for Windows

The performance of Windows systems is greatly enhanced by defragmenting or defragging the disk. A file may have many small fragments spread across the hard disk of the computer. Rearranging the fragments together in one place speeds up computer performance. The Windows OS does provide an inbuilt defrag tool, but the efficiency of the operation is improved with the use of third party tools like UltraDefrag.



UltraDefrag is an open source disk defragmenter for Windows. It improves defragmentation and speeds up computer performance to read continuous data from the disk. It uses many defragmentation algorithms to do so.

UltraDefrag has many useful features, which are shown in Figure 1.

Processing algorithms

UltraDefrag gathers the complete information about files and gaps of free space by analysing the disk. Optimising or defragging the disk depends on the selected action or algorithm.

There are three processing algorithms available for defragmentation.

The first is the *defragmentation algorithm*. It has the following phases.

- **Analysis phase:** This gathers all information about files and disk spaces.
- **Defragmentation phase:** This joins all the fragments of files into one unit. Locked or unmovable files are skipped.
- **Partial defragmentation phase:** This tries to join the rest of the fragments of files into as few units as possible.

The second is the *quick optimisation algorithm*.

- The analysis phase is the same as in the defragmentation algorithm.
- In the next phase, UltraDefrag moves all the fragments to

the beginning of the free space at the end of the disk.

- The defragmentation phase is also the same as in the defragmentation algorithm.
- In the last phase, it moves all the not-so-fragmented files to the beginning of the disk to create a large free space at the end.

The third is the *full optimisation algorithm*.

- The analysis and defragmentation phase are the same as in the defragmentation algorithm.
- In the second phase, all fragmented or unfragmented files will be moved to the beginning of the free space at the end of the disk.
- And the last phase will work the same as in the quick optimisation algorithm.

Supported file systems

UltraDefrag can defragment the following file systems:

- FAT12/16/32
- exFAT
- NTFS

There are some restrictions in UltraDefrag.

- Directories lying on FAT disks cannot be moved. This well-known FAT driver restriction cannot be fixed.
- To defragment the *\$MFT* file, we need at least Windows XP, as NTFS system drivers are not allowed to access it.